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Growth performance, efficiency, and composition of Holstein heifers fed 2 levels of nutrient intake from 44 to 350 kilograms body weight

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ABSTRACT

No published datasets describe the effects of differences in nutrient intake, initiated shortly after birth and continued beyond puberty, on the growth characteristics of Holstein heifers. The objective of this study was to evaluate the effects of 2 levels of nutrient intake on growth characteristics and composition of Holstein heifers at stages ranging from early life up to one-half of mature BW. Starting at 44 kg, heifers (n = 78) were assigned to either an elevated (E) or restricted (R) level of nutrient intake formulated to support an ADG of 950 g and 650 g, respectively. Six heifers were slaughtered at approximately 46 kg to serve as a baseline for body composition. From the remaining animals, 6 heifers per treatment were slaughtered at 50 kg increments from 100 to 350 kg. Elevated nutrient intake resulted in increased daily gains of hip height and empty body weight (EBW) and the expected reduction in age at a common BW relative to cohorts raised on a lower level of nutrient intake. The lifetime BW gains observed closely matched the intended ADG for both groups, whereas E-heifers converted feed into gain more efficiently than R-heifers until 250 kg. The E-heifers attained puberty 107 d earlier and, from that point onward, were consistently shorter than R-heifers. The proportion of gut fill in the BW was not constant during growth and was affected by level of nutrient intake after 250 kg being lower in E- than in R-heifers. A greater proportion of the blood and organs fraction and total fat mass reduced the contributions from the other body fractions to the EBW, in E- compared with R-heifers. The content of fat increased and that of water decreased in both the EBW and the gain as growth progressed. These changes in fat and water content were accentuated by the E-level of nutrient intake after 200 kg, decreasing as a result the proportion of the fat-free mass (FFM) gain in the EBW gain. The mass of final and retained protein was unaffected by level of nutrient intake. Although the protein content of the EBW and gain was reduced in E-heifers, that of the FFM gain was similar between feeding groups. The ash content of the FFM gain was reduced in E- compared with R-heifers. Changes in fat gain and deposition resulted in equivalent variations in energy retention. The elevated intake of both protein and energy improved growth and feed efficiency without affecting lean growth or increasing adiposity up to approximately 30% of the heifer's mature BW. Taking into consideration the evolving contribution of gastrointestinal fill and fat to the BW gain during development due to level of nutrient intake, these data could be used to refine current descriptions of nutrient requirements and expected growth of Holstein heifers up to 350 kg.
Key words: growth, heifer, composition of gain, nutrient, energy

INTRODUCTION

The chemical composition of preruminant Holstein heifers has received considerable research interest over the last 2 decades (Diaz et al., 2001; Tikofsky et al., 2001; Brown et al., 2005; Bartlett et al., 2006; Stamey Lanier et al., 2021). This work led to investigations of the role of nutrition and body composition on mammary development (Meyer et al., 2006a,b; Daniels et al., 2009; Geiger et al., 2016) and first-lactation milk yield and future productivity (Soberon et al., 2012; Soberon and Van Amburgh, 2013) when growth is nutritionally manipulated starting early in life. From a recent evaluation of first-lactation milk yield, changes in body composition through the growing phase relative to mature BW at calving are most likely responsible for much of the negative effect of pre- and postpubertal growth rate on first-lactation milk yield, independent of mammary development (Van Amburgh et al., 2019).

It is well understood that changes in body composition influenced by growth rate, level of energy intake, or both result from differences in visceral tissue mass and, more importantly, from the dilution of the ash, water, and protein by fat (Reid et al., 1968; Garrett, 1987; McLeod

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

and Baldwin, 2000). This is because the chemical composition of the fat-free matter (**FFM**) gain is considered largely unresponsive to BW or the level of nutrient intake (Fortin et al., 1980; Garrett, 1987; Waldo et al., 1997). The retention of body protein relative to body fat decreases with increasing BW and increasing energy intake, diluting the protein composition of the empty body. It is assumed that the rate of protein retention becomes limiting as energy intake increases beyond a particular level, and therefore, energy not retained as protein is retained as fat (Byers, 1982).

Except for data collected at Cornell University (Ithaca, NY) in the 1950s and 1960s (Fortin et al., 1980), little serial slaughter data exist describing the composition of retained tissue in the growing Holstein heifer beyond weaning and through puberty. There are data evaluating the effects of diet, nutrient intake or both on the composition of growth between approximately 100 kg (Moallem et al., 2004b) or 180 kg (Waldo et al., 1997) BW and puberty in Holstein heifers; however, there was no consideration of the effect of preweaning nutrition or the adaptations to the ruminant state after weaning on the composition of the heifers in either of the datasets. The data of Waldo et al. (1997) suggested that Holstein heifers had less adipose tissue than previously observed and that this might be due to changes in mature size associated with selection for milk yield. However, the effect of the level of nutrient intake initiated shortly after birth and sustained beyond puberty on growth composition has yet to be thoroughly investigated.

Although the rapid growth of prepubertal heifers produced by increased energy intake has been associated with undesirable outcomes, that obtained by increased protein and energy intake has decreased age at puberty without detrimental effect on stature or mammary gland development (Radcliff et al., 1997) and reduced age at first calving without affecting milk production (Van Amburgh et al., 1998).

The objective of this study was to describe the effects of 2 levels of nutrient intake, initiated early in life, on growth characteristics, efficiency and composition in the Holstein heifer at different stages of development, through puberty and up to one-half of mature BW. An additional objective was to meet the energy requirements to achieve target growth rates while ensuring protein would not be the most limiting nutrient.

MATERIALS AND METHODS

Animal Management

The Cornell University Animal Care and Use Committee approved all procedures used in this study. The overall study design, described in Meyer (2005), was co-developed to evaluate mammary development throughout prepubertal growth while providing a robust dataset for body composition analysis. To do this, heifers were slaughtered every 50 kg from 100 to 350 kg BW to evaluate mammary epithelial cell proliferation and total parenchymal accumulation, and these data were previously published (Meyer et al., 2006a,b). To make use of the entire animal, body composition was measured at the same intervals to assess body composition changes due to changes in BW and nutrient intake, and this approach was similar to a study by Fortin et al. (1980) where both Holstein and Angus cattle of different sexes were slaughtered for evaluation and comparison of chemical composition. Thus, the current study design employs a modified serial slaughter technique with multiple intervals, rather than baseline and endpoint composition. Additionally, because dairy replacement heifers are not typically grown at rates of gain similar to finishing steers, the diet composition was modified throughout the study to maintain the target rates of gain, ensuring that protein was not limiting growth. This approach also differs from more traditional serial slaughter studies. This study was conducted from October 2001 to July 2003.

Seventy-eight Holstein heifers were purchased within 1 wk of birth from dairy farms near Ithaca, New York, from October 2001 to June 2002. The mature size of the cattle on 4 of the 5 farms was measured using measuring tapes on cattle in the middle of the fourth lactation. The average mature BW identified among the 4 farms was 657 ± 117 kg. After arrival at the research farm (44.2 kg, 9.9 d old), heifers were randomly assigned to either an elevated (**E**) or restricted (**R**) level of nutrient intake and were fed to achieve 650 g (R) or 950 g (E) daily gain.

Before weaning, all heifers were fed twice daily at 0630 and 1800 h. The E-heifers received a 28% CP (all milk protein) and 20% fat milk replacer on a DM basis (Cows Match, Land O'Lakes Animal Milk, Arden Hills, MN) at 0.32 Mcal intake energy per kilogram of metabolic body weight ($BW^{0.75}$). In comparison, R-heifers received a 22% CP (all milk protein) and 20% fat milk replacer on a DM basis (Land O'Lakes Animal Milk, Arden Hills, MN) at 0.20 Mcal intake energy per kg of $BW^{0.75}$ (Table 1). Heifers were weighed once weekly, and the amount of milk replacers offered was adjusted accordingly to maintain the target growth rates. Weaning was initiated after approximately 42 d of treatment by reducing the amount of milk provided to 50% the preweaning amount for 7 d and then removing all milk replacer by approximately 50 d. To achieve a BW at weaning as similar as possible between treatments, the R-heifers were offered milk for 7 d longer than the E-heifers. Calves had access to water throughout the entire treatment period. A commercially available 26% CP textured starter was offered starting at approximately 1 wk on study until 2 wk after weaning

Table 1. Milk replacer and calf starter DM and nutrient composition

	Diet		
Nutrient, % of DM	R-milk replacer[1]	E-milk replacer[2]	Calf starter[3]
DM, %	96.0	96.0	90.8
CP	22.0	29.3	26.7
Fat	21.2	18.9	4.2
Ca	0.95	0.99	1.41
P	1.04	0.97	0.65
NDF	NA[4]	NA	18.3
ADF	NA	NA	10.2

[1]Milk replacer offered to heifers assigned to the restricted plane of nutrition (Land O'Lakes Inc., St. Paul, MN).

[2]Milk replacer offered to heifers assigned to the elevated plane of nutrition (Land O'Lakes Inc., St. Paul, MN).

[3]Textured calf starter (Agway Inc., Syracuse, NY).

[4]Not applicable.

Table 2. Total mixed rations DM and nutrient composition

	TMR		
Item	1	2	3
Ingredient, % of DM			
Alfalfa hay	10.3	—	—
Alfalfa silage	15.1	31.8	31.8
Corn silage	13.8	31.8	31.8
Soybean meal	—	—	4.2
Heat-treated soybean meal	10.3	18.0	8.4
Starter grain mix[1]	50.3	—	—
High-moisture corn	—	15.9	23.1
Monensin mix[2]	—	1.6	1.7
Vitamin E[3]	0.12	0.10	—
Mineral mix[4]	0.12	0.9	1.4
Nutrient, % of DM			
DM, %	60.3	45.3	46.7
CP	24.3	19.0	17.2
ADF	18.1	21.1	20.2
NDF	35.5	34.0	31.4
NSC[5]	34.3	35.9	40.0
Fat	1.91	3.93	3.45
Ash	8.48	7.41	7.21
ME,[6] Mcal/kg DM	2.43	2.45	2.47

[1]Calf starter described in Table 1.

[2]Contained 1,760 mg of monensin (Elanco Animal Health, Greenfield, IN) per kilogram of DM.

[3]Supplement contained 44 IU/g vitamin E.

[4]Supplement contained 58.6% corn meal, 30.5% NaCl, 4.7% calcium sulfate, 1.9% calcium carbonate, 1.3% manganese sulfate, 1.1% zinc sulfate, 0.55% ferrous sulfate, 0.53% vitamin E (500 IU/kg), 0.33% copper sulfate, 0.21% magnesium oxide, 0.20% selenium (1.0%), 0.034% vitamin A, 0.022% vitamin D, 0.004% cobalt carbonate, and 0.004% iodine.

[5]Nonstructural carbohydrates, % NSC = 100 − (% ash + % CP + % NDF + % ether extract).

[6]Calculated using the Cornell Net Carbohydrate and Protein System v6.5 (Van Amburgh et al., 2015).

(Table 1; Agway Inc., Syracuse, NY). These diets met the CP requirements for the expected ADG, as calculated by Van Amburgh and Drackley (2005), and were further confirmed by the NASEM (2021) recommendations.

To maintain the target growth rates by stage of maturity, 3 TMR common to both treatments were formulated using the Cornell Net Carbohydrate and Protein System (**CNCPS** v5, Fox et al., 2004) such that predicted ME and MP allowable daily gains were similar within the BW range in which they were offered and across the 2 targeted rates of gain. For the 2 wk following weaning (wk 8 and 9), all heifers were offered the starter to ensure adequate feed intake to meet the growth objectives. From 10 to 13 wk of age, heifers received TMR 1 (Table 2), after which they were switched to TMR 2 (Table 2). The heifers were fed TMR 2 until they reached approximately 200 kg BW. From 200 kg BW until slaughter, TMR 3 (Table 2) was offered to the remaining heifers. The amount of feed offered was adjusted to achieve the target growth rate of gain using the CNCPS to formulate the diet. The TMR were formulated to facilitate the transition of heifers from a starter to a forage-based diet (TMR 1) and to meet their nutrient requirements during a BW gap of more than 200 kg (TMR 2 and 3). According to NASEM (2021), these diets provided protein adequate to sustain 980 g of daily gain for all BW ranges, which supports that protein was not limiting growth in this study. From the initiation of treatment to 150 kg BW, all heifers were housed and fed in individual pens within a greenhouse structure at the Cornell Dairy Teaching and Research Center (Harford, NY). Upon reaching 150 kg BW, heifers were moved to a freestall barn, group housed and individually fed via the Calan gate system (American Calan, Northwood, NH) at the Teaching and Research Center.

Throughout the study, heifers were weighed, and hip height (**HH**) was measured once per week. The BW and HH rates of gain were calculated as the slopes of the regressions of body measurements on age for each heifer, then averaged by treatment. Once heifers reached 225 kg, blood was collected twice weekly via jugular venipuncture, and plasma progesterone concentration was determined (Coat-A-Count progesterone radioimmunoassay, Diagnostic Products Corporation, Los Angeles, CA). Progesterone concentrations above 1 ng/mL were interpreted as indicating that the heifer had a functional corpus luteum and was therefore pubertal (Meyer et al., 2006a,b).

Feed Ingredient Sampling and Analysis

Diet samples were collected daily, stored at −20°C, and composited by week. These composites were used to determine TMR DM by drying at 105°C for 24 h. Dry matter values from these weekly composites were used to calculate daily DM intake. All diet ingredients were sampled weekly and composited by month. These composites were dried at 105°C to a constant weight and retained for proximate analysis. Acid detergent fiber (AOAC, 1990), ash (AOAC, 1990), crude fat (AOAC, 1990), N (AOAC

International, 2000), and NDF (Mertens et al., 2002) were performed on all composites. In all cases, crude protein was calculated as N × 6.25. Nonstructural carbohydrate was estimated by difference [%NSC = 100 − (%ash + %CP + %NDF + %crude fat)]. Starch and sugar analyses were not conducted in this study.

Animal Slaughter and Tissue Collection

All heifers were purchased within 1 wk of birth, transported to the farm and assigned to treatment. For the heifers assigned to the baseline composition group, they were held at least 1 wk after arrival and fed the respective treatment diet until they reached approximately 45 to 46 kg BW. The 6 heifers (3 per treatment) were slaughtered at 45.8 ± 1.3 kg at approximately 21 d of age to determine baseline body composition at the initiation of treatment. The remaining heifers were slaughtered at one of the following target BW: 100, 150, 200, 250, 300, or 350 kg (6 heifers per treatment per slaughter weight). Animal slaughter was conducted at the Cornell University abattoir (Ithaca, NY) by stunning with a captive bolt followed by exsanguination.

Heifers were weighed before transport to the abattoir (a 30-min ride) and again before slaughter on 2 certified scales. The difference between pre- and post-transport BW was defined as transport shrink. Body tissue of slaughtered animals was divided into 4 components for compositional analysis. These components consisted of (1) head, hide, feet, and tail (**HHFT**); (2) mammary gland (**MG**); (3) blood and organs excluding the MG (**BO–MG**); and (4) carcass (**CA**). The front and back legs were removed at the carpal and tarsal joints, respectively, and combined with the hide and head, which were removed between the skull and the atlas. These tissues were weighed together and comprised the HHFT fraction. Blood from exsanguination was collected in a plastic barrel and weighed. All organs, including the gastrointestinal tract (**GIT**), were removed and weighed. The gastrointestinal tract was cleaned of digesta, rinsed with water, and reweighed for determination of GIT fill and empty BW (**EBW**). All organs (excluding the MG) were combined with the collected blood to represent the BO–MG fraction. Finally, the carcass was weighed and split longitudinally. Each half was reweighed, and the right half of the carcass was retained as the CA fraction. The chemical composition of the CA (representing half of the carcass) was multiplied by the weight of the entire carcass to establish whole carcass values. The sum of all tissue component weights (HHFT, MG, BO–MG, and whole CA) represented the heifers' EBW.

These tissue components were stored at −20°C for homogenization later. After partial thawing at room temperature, each component (HHFT, BO–MG, and CA) was individually ground (Model 801B Grinder, Autio Co., Astoria, OR). For the first pass through the grinder, an 18-mm screen was used, followed by 6 consecutive passes using an 8-mm screen. Following the final pass through the grinder, 1-kg grab samples were collected, bulked, subsampled, and stored at −20°C for later analysis. The MG was processed as previously described by Meyer et al. (2006 a,b).

Component samples were freeze-dried for the initial determination of DM. Final DM of component samples was determined by drying at 105°C for 72 h. All values for the final DM were calculated as the freeze-dried DM multiplied by the oven DM. Freeze-dried samples were ground with dry ice through a 2-mm screen (Wiley Mill, Arthur H. Thomas, Philadelphia, PA). Ground component samples were analyzed for crude fat, N, and ash by the same methods described above for feed ingredients. In all cases, CP was calculated as N × 6.25. Tissue energy (**TE**) of the EBW and its fractions were calculated based on fat and protein contents and the respective combustion values reported by Blaxter and Rook (1953; 9.367 kcal/kg of fat and 5.322 kcal/kg of protein).

The GIT fill was calculated as the difference between live BW pretransport and EBW after emptying the GIT. Because baseline animals were of similar BW as the other heifers at the initiation of the study, the composition of the baseline animals' EBW (both proportion of HHFT, MG, BO–MG, and CA and their chemical composition) was assumed to be similar to that of all the other heifers at the initiation of the study. Based on this assumption, for each of the 72 heifers assigned to a level of nutrient intake and slaughter BW, the initial EBW and empty body composition were calculated using their live BW at the initiation of treatment and the data from the baseline slaughter group of their respective treatment. For the present study, the MG weight and composition were added to those of the BO–MG to represent the entire blood and organs (**BO**) fraction.

Statistical Analysis

Measurements taken before slaughter (DMI, days on feed, body measurements at the beginning of the study and weaning, preweaning growth, and measurements related to the onset of puberty) were analyzed as a completely randomized design using a mixed-effect model that included the fixed effect of treatment (level of nutrient intake) and the random effect of the month of enrollment. The measurements taken at slaughter (days on treatment, BW, HH, and fraction and EBW composition) and those deriving from them were analyzed as a completely randomized design, including the fixed effects of treatment, slaughter weight, and their interaction and the random effect of the month of enrollment. Heifer was the experimental unit, and slaughter weight was treated as a categorical variable.

Analyses were performed using R (v. 4.3.1; R Core Team, 2023) with the "lme4" package (Bates et al., 2015) "lmer ()" function. The "emmeans" package (Lenth et al., 2019) was used to calculate the least squares means of the main effects and their interaction. To reduce the chances of overlooking a biologically significant interaction of the main effects, the interaction means were compared within slaughter weight groups when the F-test suggested a potential interaction ($P \leq 0.10$) by pairwise comparisons using least squares means with Tukey's adjustment to correct for multiple comparisons. All reported mean values are least squares means, while standard error is the variation parameter.

To evaluate the relationships of EBW with empty body fat (**EBF**) or empty body protein (**EBP**), and between FFM and the different chemical components of the body, the allometric function, $Y = aX^b$, of Huxley (1932) was employed as done by Fortin et al. (1980). In this equation, "b" represents the growth rate of Y (EBF or EBP) relative to that of X (EBW or FFM). These regressions were fitted using the 'nlme' function from the 'nlme' package (V3.1–162; Pinheiro and Bates, 2020). Regressions were performed as mixed-effect models, where fixed effects varied by equation, and the random effect of the month of enrollment was included. Comparisons of the "b" coefficients within treatments were made using the 'pairs' function from the 'emmeans' package. To compare "b" coefficients of the different component's regression with FFM, a Z-test was performed using the estimated parameters and their respective SE. Significance was declared at $P \leq 0.05$. The daily retained energy (**RE**) as protein and fat was regressed against total daily RE using the "lm ()" function.

RESULTS

Composition of Baseline Animals

To serve as baseline data for comparison of growth and composition changes, 3 heifers per treatment were slaughtered at a BW similar to the BW at which the remaining heifers started treatment (Table 3). The average BW and composition of the baseline heifers from each treatment are reported in Table 4.

DMI and Feeding Length

As intended, the DMI of milk replacers, starters, and the 3 TMR was greater for the E-heifers than R-heifers ($P < 0.01$, Table 5), as the amount of feed offered was adjusted weekly to maintain the target growth rates. The R-heifers were fed longer than E-heifers ($P < 0.03$, Table 5) to reach a common slaughter BW. The more prolonged milk feeding phase, about 7 d, for the R-heifers was intended to ensure both groups of heifers finished at a similar BW at weaning. As indicated in Table 5, all heifers received milk replacer, starter, and TMR 1 during the treatment period. As heifers achieved the 100 kg slaughter BW during the TMR 1 feeding period, only the remaining animals received the subsequent TMR. Due to temporary space constraints in the free stall barn, 2 E-heifers assigned to the 200 kg slaughter BW were transferred to another pen where they received the TMR 3 for less than 13 d before slaughter at a DMI to maintain the target growth rate.

Table 3. Age and body measurements at the beginning of the experiment and at weaning for heifers fed 2 levels of nutrient intake, elevated (E) or restricted (R), from early life

	Treatment			
Item	E	R	SEM[1]	P-value
Initial age, d	10.4	9.4	0.74	0.30
Initial BW, kg	43.7	43.9	1.19	0.85
Initial HH[2]	82.7	81.9	3.37	0.24
BW at weaning,[3] kg	83.9[a]	74.0[b]	0.99	<0.01
HH at weaning, cm	93.9[a]	92.4[b]	0.42	<0.01
Preweaning BW gain, g/d	882[a]	590[b]	28.20	<0.01
Preweaning HH gain, cm/d	0.26[a]	0.21[b]	0.021	0.01

[a,b]Within rows, means with uncommon superscripts are different at $P < 0.05$.

[1]Largest SEM is shown.

[2]Hip height.

[3]The days until weaning correspond to the days on milk replacer reported in Table 5.

Table 4. Body weight and composition of the baseline heifers fed nutrients at an elevated (E) or restricted (R) level

	Treatment[1]	
Item	E	R
Pretransport BW[2]	45.6 ± 0.7	46.1 ± 1.6
EBW,[3] % pretransport BW	95.1 ± 1.8	93.8 ± 2.1
BO,[4] % EBW	19.9 ± 0.6	19.6 ± 0.7
Carcass, % EBW	60.1 ± 0.6	60.9 ± 0.9
HHFT,[5] % EBW	20.0 ± 0.1	19.5 ± 0.2
CP, % of EBW	16.9 ± 0.1	16.5 ± 0.2
Fat, % of EBW	6.1 ± 0.8	6.2 ± 0.6
Ash, % of EBW	3.8 ± 0.5	3.9 ± 0.2
H_2O, % of EBW	73.2 ± 0.9	73.4 ± 0.8

[1]Values are means ± SD (n = 3).

[2]Live BW at the Cornell University farm (Harford, NY) before transport to the abattoir (a 30-min ride).

[3]Empty BW.

[4]BO = blood and organs.

[5]HHFT = head, hide, feet, and tail.

Table 5. Dry matter intake and feeding days of milk replacers, starter grain, and TMR diets[1] fed to Holstein heifers raised on an elevated (E) or restricted (R) nutrient intake level up to 6 slaughter weights

	DMI, kg/d			Days on feed			Animals	
Feed	E	R	SEM	E	R	SEM	E	R
MR[2]	1.20[a]	0.76[b]	0.016	45.9[b]	51.5[a]	2.07	36	36
Starter[3]	1.04[a]	0.71[b]	0.049	57.3[b]	63.4[a]	4.45	36	36
TMR 1	2.84[a]	2.17[b]	0.070	16.7[b]	19.1[a]	2.86	36	36
TMR 2	4.19[a]	2.97[b]	0.140	58.8[b]	118.9[a]	22.8	30	34
TMR 3	6.72[a]	4.59[b]	0.268	59.3[b]	93.8[a]	23.4	20	18

[a,b]Within variables, means with uncommon superscripts differ at $P < 0.05$. The effect of treatment was significant ($P < 0.03$) for the DMI and days on each feed.

[1]TMR 1 was offered to all heifers from 10 wk of age to 13 wk of age. TMR 2 was provided from 13 wk of age until they reached approximately 200 kg BW. TMR 3 was provided to all heifers from 200 to 350 kg BW. Ingredient and nutrient composition are provided in Table 2.

[2]MR = milk replacer; each treatment group received its respective MR (E and R), as indicated in Table 1.

[3]The starter was offered from approximately the second week of the study until 2 wk after weaning (wk 8 and 9). The nutrient composition is provided in Table 1.

Body Growth and Components

At the initiation of treatment, the means of BW, age, and HH (44.2 kg, 9.9 d, and 82.4 cm, respectively) were similar among treatments (Table 3). Despite the efforts to bring both groups to similar BW at weaning, E-heifers were both heavier and taller at weaning ($P < 0.01$, Table 6) than R-heifers. This difference in BW and HH at weaning was achieved through greater ADG and HH gain in the E-heifers than in the R-heifers.

The E-heifers required fewer days on treatment than R-heifers ($P < 0.01$) to achieve the target slaughter weights, requiring 129 d less to attain 350 kg (Table 6). As expected, the DMI as percentage of BW, from all feeds combined, was greater for E-heifers compared with their R- counterparts irrespective of BW at slaughter. However, heifers slaughtered at 100 kg had a lower DMI respective to their BW, compared with those slaughtered at subsequent BW, regardless of treatment. On the day of slaughter, heifers were weighed before leaving the farm and once again upon arrival at the abattoir. The transport shrink expressed as a percent of pretransport BW was not influenced by treatment or BW (3.96% BW, $P = 0.24$ and 0.49, respectively). Live BW pre-transport and at slaughter were not different between E- and R-heifers and closely matched the target slaughter weights. However, E-heifers had a heavier EBW and greater EBW as a percent of live BW than R-heifers at 300 and 350 kg ($P < 0.01$), and this was attributed to a greater amount of GIT fill, absolute and relative to live BW, in R- than E-heifers ($P < 0.01$, Table 6 and Supplemental Figures S1A and S1B, see Notes), as indicated by the treatment and slaughter weight interaction. Although GIT fill as a proportion of BW remained relatively constant across slaughter BW, it was greater at 150 kg ($P < 0.02$) regardless of level of nutrient intake. Consequently, the proportion of EBW in the live BW was lower at 150 kg compared with the other slaughter BW ($P < 0.02$). Estimated initial EBW was not different between the 2 treatments ($P = 0.94$), however, it tended to vary ($P = 0.06$) among calves assigned at slaughter BW most likely due to the inherent variation from the population calves were drawn from (Table 6). The E-heifers were shorter than the R-heifers at 150, 300, and 350 kg as indicated by the interaction found for HH at slaughter between the level of nutrient intake and slaughter weight (Table 6).

Lifetime live BW, EBW, and HH daily gain followed the same trend as the preweaning gains (Table 6). The E-heifers had average lifetime daily live BW and EBW gains (**EBG**) of 930 and 730 g/d, respectively. The R-heifers grew significantly slower ($P < 0.01$) with lifetime ADG and EBG of 663 and 501 g/d, respectively. After separating lifetime ADG and EBW daily gain by the treatment and slaughter weight, it became apparent that ADG was not constant throughout the heifers' development. Indeed, both ADG and EBW daily gain increased as BW increased for heifers on both treatments (Table 6); however, the proportion of EBG in the ADG was lower in R- than in E-heifers at 300 and 350 kg (Supplemental Figure S1B). This proportion of EBG in the ADG was lower at 100 and 150 kg ($P < 0.02$) compared with those observed in subsequent slaughter BW. Similarly, the E-heifers had a faster lifetime HH daily gain than R- counterparts ($P < 0.01$) at the different slaughter weights, but this gain decreased as BW increased for both treatments. Feed efficiency, calculated as the ratio between the EBG and the DMI, both expressed in kilograms per day, decreased as growth progressed, and was higher in E-heifers compared with R-heifers until 250 kg and similar between treatments thereafter.

The chemical composition of the empty body is in Table 7. The total amount of EBP, ash, and water was not

Table 6. Effect of nutrient intake level on days on feed, DMI, growth, and body components among 6 slaughter weights in Holstein heifers raised at an elevated (E) or restricted (R) nutrient intake level

	Treatment														
	100 kg		150 kg		200 kg		250 kg		300 kg		350 kg		*P*-value		
Item	E	R	E	R	E	R	E	R	E	R	E	R	TRT[1]	BW[2]	Int.[3]
Initial EBW,[4] kg	37.9	37.7	41.0	41.2	43.1	40.1	44.4	44.2	42.3	42.9	42.1	44.1	0.94	0.06	0.76
Days on treatment	74.0[b]	103.6[a]	118.3[b]	176.6[a]	168.6[b]	251.6[a]	221.5[b]	305.3[a]	271.1[b]	369.2[a]	328.6[b]	457.2[a]	<0.01	<0.01	<0.01
DMI, % BW	1.50[a]	1.35[b]	1.99[a]	1.61[b]	2.07[a]	1.70[b]	2.16[a]	1.68[b]	2.28[a]	1.62[b]	2.32[a]	1.59[b]	<0.01	<0.01	<0.01
Pretransport BW[5]	102.5	104.8	151.9	151.7	203.8	204.7	255.1	256.0	303.8	299.8	359.5	361.0	0.84	<0.01	0.69
BW[6] at slaughter, kg	98.4	100.3	143.4	145.7	194.2	196.7	246.6	246.8	291.2	292.1	347.2	348.0	0.34	<0.01	0.97
HH[7] at slaughter, cm	99.8	98.7	104.9[b]	109.9[a]	116.8	116.8	125.1	125.2	126.3[b]	129.5[a]	132.7[b]	135.4[a]	<0.01	<0.01	0.01
Gut fill,[8] kg	18.0	20.1	32.4	34.1	38.0	37.0	43.7	47.8	47.6[b]	57.6[a]	50.4[b]	68.8[a]	<0.01	<0.01	<0.01
EBW at slaughter, kg	84.5	84.5	119.2	117.4	165.5	167.5	211.4	208.1	256.6[a]	242.6[b]	309.5[a]	292.6[b]	<0.01	<0.01	<0.01
ADG,[9] g/d	848	616	923	629	944	665	951	697	958	686	958	684	<0.01	<0.01	0.66
EBG,[10] g/d	639	455	664	446	730	522	758	543	777	525	799	528	<0.01	<0.01	0.16
HH gain,[11] cm/d	0.215	0.168	0.174	0.154	0.181	0.136	0.181	0.126	0.171	0.132	0.155	0.117	<0.01	<0.01	0.95
Feed efficiency[12]	0.61[a]	0.46[b]	0.34[a]	0.28[b]	0.28[a]	0.24[b]	0.23[a]	0.21[b]	0.21	0.20	0.18	0.17	<0.01	<0.01	<0.01

[a,b]Means with uncommon superscripts within a common slaughter weight are different at $P < 0.05$.

[1]Treatment, R = restricted nutrient intake, E = elevated nutrient intake.

[2]Slaughter weight.

[3]Interaction of TRT and SW.

[4]Empty BW. Initial EBW was calculated as 95.1% and 93.8% of the initial live BW for the E- and R-heifers, respectively.

[5]Live BW at the Cornell University farm before transport to the abattoir (a 30-min ride).

[6]Live BW upon arrival at the Cornell abattoir.

[7]Hip height.

[8]Digestive tract contents found by difference between live BW pretransport and EBW after emptying the gastrointestinal tract.

[9]Lifetime live BW ADG from initiation of treatment to slaughter.

[10]EBW ADG.

[11]Lifetime HH daily gain from initiation of treatment to slaughter.

[12]EBG/DMI, both in kg/d.

Table 7. Effect of nutrient intake level on whole body composition among 6 slaughter weights in Holstein heifers raised at an elevated (E) or restricted (R) level of nutrient intake

Item	Treatment												*P*-value		
	100 kg		150 kg		200 kg		250 kg		300 kg		350 kg				
	E	R	E	R	E	R	E	R	E	R	E	R	TRT[1]	BW[2]	Int.[3]
Fat, kg	8.4	7.3	13.6	13.9	20.4	22.6	34.6	30.4	53.5[a]	38.4[b]	70.1[a]	52.3[b]	<0.01	<0.01	<0.01
CP, kg	15.3	15.7	21.2	20.9	29.4	29.6	34.6	34.4	38.1	37.6	46.2	44.9	0.29	<0.01	0.70
Ash, kg	3.4	3.7	5.0	5.4	7.1	7.1	8.4	8.9	9.4	9.9	12.9	12.7	0.18	<0.01	0.91
H_2O, kg	58.2	59.2	80.9	78.7	110.2	109.6	134.4	135.1	151.2	152.5	176.1	178.4	0.67	<0.01	0.73
FFM,[4] kg	76.7	78.3	107.0	104.9	146.6	146.1	177.4	178.3	199.3	200.4	235.6	236.4	0.79	<0.01	0.94
TE,[5] Mcal	156	148	237	239	344	366	507	466	716[a]	572[b]	914[a]	741[b]	<0.01	<0.01	<0.01
Fat, % of EBW[6]	10.0	8.6	11.3	11.8	12.3	13.5	16.4[a]	14.6[b]	20.8[a]	15.8[b]	22.6[a]	17.8[b]	<0.01	<0.01	<0.01
CP, % of FFM	19.2	19.2	19.5	19.6	19.8	20.0	19.5	19.2	19.8	19.5	20.2	19.6	0.15	0.04	0.19
Ash, % of FFM	4.5	4.8	4.6	5.1	4.8	4.9	4.7	5.0	4.7	4.9	5.4	5.4	0.06	0.03	0.75
H_2O, % of FFM	76.5	76.1	75.9	75.3	75.4	75.1	75.8	75.8	75.3	75.5	74.3	75.0	0.67	<0.01	0.15
TE, Mcal/kg EBW	1.85	1.74	1.98	2.02	2.08	2.18	2.40[a]	2.24[b]	2.78[a]	2.36[b]	2.94[a]	2.52[b]	<0.01	<0.01	<0.01

[a,b]Means with uncommon superscripts within a common slaughter weight are different at $P < 0.05$.

[1]Treatment, R = restricted nutrient intake, E = elevated nutrient intake.

[2]Slaughter weight.

[3]Interaction of TRT and SW.

[4]Fat-free matter.

[5]Calculated tissue energy based on fat and protein mass, and their respective combustion values reported by Blaxter and Rook (1953; 9,367 kcal/kg of fat and 5,322 kcal/kg of protein).

[6]Empty BW.

influenced by treatment ($P > 0.17$). The significant interaction between the level of nutrient intake and slaughter BW indicated a greater amount of total fat in the empty body and fat as a percentage of EBW at the 2 and 3 heaviest slaughter weights, respectively, in E- compared with R-heifers. This interaction also revealed that water as a percentage of EBW was lower at the heaviest 3 slaughter weights in the E- compared with the R-heifers (data not shown). Protein as a percentage of EBW was lower in E than in the R-heifers (16.6% vs. 16.8%, respectively, $P = 0.01$) and was also affected by slaughter BW: remaining constant at 17% to 200 kg and decreasing to about 16% at 250 kg to stay unchanged until 350 kg. Ash as a percentage of EBW was lower in E- than in R-heifers (4.0% vs. 4.3%, respectively; $P < 0.01$), regardless of BW.

Fat-free matter consists of the empty body's ash, water, and protein and represents the retained tissue independent of the dilution effect of adipose tissue, related to growth rate and relative mature BW. The amount of FFM and its water and protein content were unaffected by treatment ($P > 0.15$, Table 7), but E-heifers tended to have a lower ash concentration in the FFM than their R-heifer counterparts (4.8% vs. 5.0%, respectively, $P = 0.06$). The FFM content of water decreased, whereas that of protein and ash increased, as slaughter BW increased ($P < 0.03$, Table 7).

An interaction between nutrient intake and slaughter was observed for both the calculated total TE and TE per kilogram of EBW, following changes in fat amount and fat percentage in EBW (Table 7). The calculated TE was higher at the last 2 slaughter BW for E- than R-heifers ($P < 0.01$), whereas the calculated TE per kilograms of EBW was greater for E- than R-heifers ($P < 0.03$) at the heaviest 3 slaughter weights.

The relationship between EBW, EBF, and EBP was characterized using the allometric equation, $Y = aX^b$ (Figure 1). The 2 curves confirm that E-heifers had more EBF as a fraction of EBW than R-heifers as EBW increased, and the equations describing this relationship are as follows: E-heifers, $EBF = 0.002 \times EBW^{1.82 \pm 0.06}$, adjusted (**Adj.**) $R^2 = 0.94$; R-heifers, $EBF = 0.010 \times EBW^{1.50 \pm 0.07}$, Adj. $R^2 = 0.96$ (Figure 1A). The (b) coefficient, representing the fat deposition relative to that of EBW, was greater for E- than for R-heifers. ($P < 0.01$). The relationship between EBW and total EBP indicated a similar amount of tissue protein as a function of EBW for both feeding levels, and this is made apparent by the similarity between the resulting equations: E-heifers, $EBP = 0.261 \times EBW^{0.91 \pm 0.02}$, Adj. $R^2 = 0.98$; R-heifers, $EBP = 0.225 \times EBW^{0.94 \pm 0.02}$, Adj. $R^2 = 0.98$ (Figure 1B). Protein accretion relative to EBW was similar between the 2 treatments ($P = 0.30$).

The contribution of the different body fractions to the empty BW and composition is presented in Supplemental

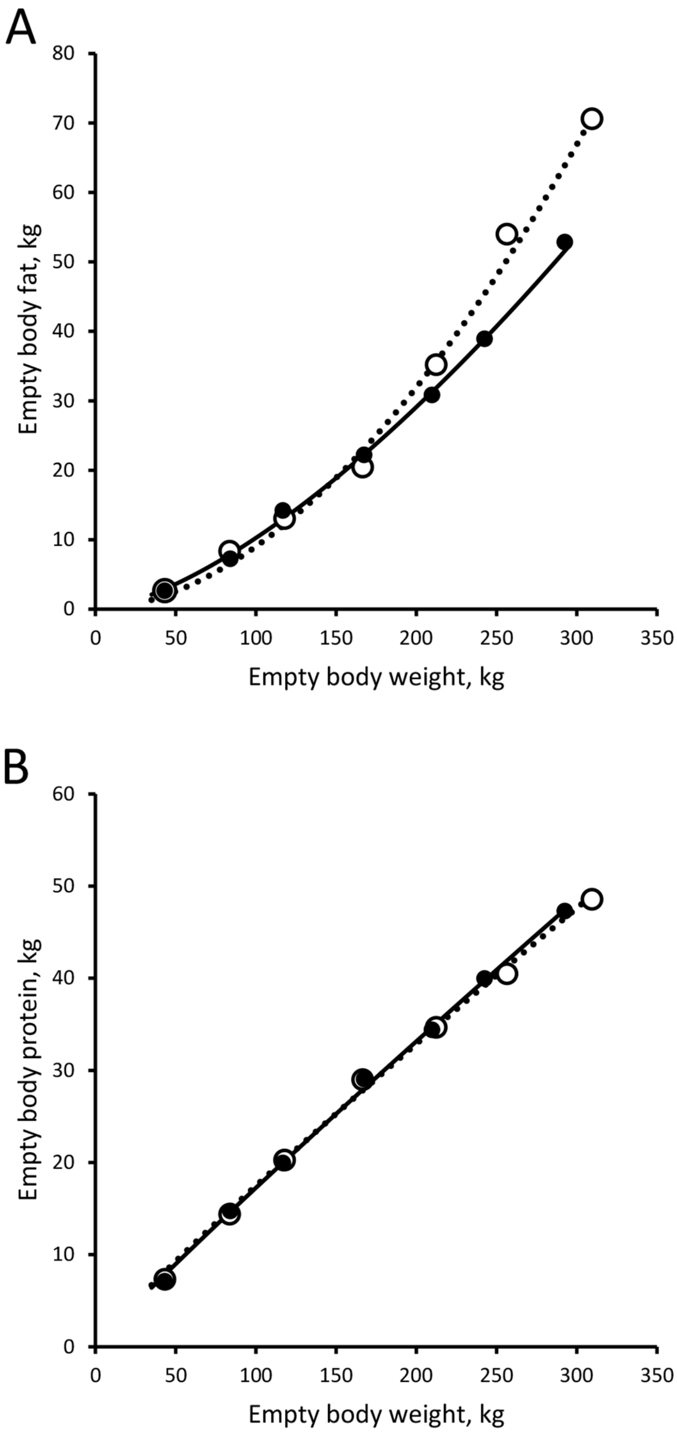

Figure 1. Relationship of empty BW (EBW) with empty body fat (EBF, panel A) or empty body protein (EBP, panel B) of Holstein heifers raised on an elevated (E, open circles) or restricted (R, solid circles) level of nutrient intake starting at 46 kg BW. E-heifers (dotted lines): EBF = 0.002 × $EBW^{1.82 \pm 0.06}$, $P < 0.01$, Adj. $R^2 = 0.94$; EBP = 0.261 × $EBW^{0.91 \pm 0.02}$, $P < 0.01$, Adj. $R^2 = 0.98$; R-heifers (solid lines): EBF = 0.010 × $EBW^{1.50 \pm 0.07}$, $P < 0.01$, Adj. $R^2 = 0.96$; EBP = 0.225 × $EBW^{0.94 \pm 0.02}$, $P < 0.01$, Adj. $R^2 = 0.98$. The EBF accretion rates differed between levels of nutrient intake ($P < 0.01$), but those of EBP did not differ between levels of nutrient intake ($P = 0.30$).

Table S1 (see Notes). The proportion of BO in the EBW was affected by feeding level, being greater for E- than R-heifers (23.4% vs. 21.7%, respectively, $P < 0.01$) regardless of slaughter weight. Therefore, this decreased the contribution of CA (59.9% vs. 61.0%, $P < 0.01$) and HHFT (16.7% vs. 17.3%, $P < 0.01$) in the EBW of E-heifers compared with their restricted counterparts. In addition, the contribution of these fractions to EBW was altered by slaughter weight, increasing and decreasing from the initial to the final weights for CA and HHTF, respectively. The chemical composition of the different body fractions showed a pattern similar to that observed for the EBW, with fat content increasing as heifers grew and nutrient intake increased, particularly at the heaviest BW, thereby decreasing the percentage of FFM (data not shown). The level of nutrient intake affected the fat contributions to the EBF ($P < 0.01$), with E-heifers having a greater contribution of BO fat (28.3% vs. 23.7%, respectively, $P < 0.01$) and a lower proportion of CA fat (59.0% vs. 61.8%, $P < 0.01$) than R-heifers. Except for heifers slaughtered at 200 and 350 kg, the contribution of fat from HHTF to EBF was significantly lower in E- than R-heifers. The percentage of EBF coming from BO fat increased with slaughter BW, whereas that coming from CA fat remained relatively constant except for a significant reduction at 200 kg. An interaction between level of nutrient intake and slaughter BW indicated that E-heifers had a greater contribution of BO protein to the EBP than the R-heifers at 150, 200, and 300 kg. The percentage of EBP coming from CA protein was not affected by treatment or slaughter BW ($P > 0.21$), whereas that coming from HHFT protein increased from the lightest to the heaviest BW and was lower for E- than R-heifers (22.3% vs. 23.1%, $P < 0.01$). The contribution of TE from the different body fractions to the empty body TE followed the same pattern as those described for the fractions' fat to the EBF.

Composition of Retained Body Tissue

Retained body tissue is defined as the differences between the final and initial EBW, TE, and composition. The composition of the retained EBW is in Table 8. Despite being slaughtered at the same BW, E-heifers retained more EBW at the 2 heaviest weights than R-heifers. Patterns and differences observed in the composition of the retained body tissue were similar to those described for the chemical composition of the EBW. The retention of protein, ash, and water was not affected by feeding level ($P > 0.20$), whereas fat was greater for E- than R-heifers in the 2 heaviest slaughter BW. Daily retention rates for all chemical components were greater in E- than in R- heifers. The significant interaction between the level of nutrient intake and slaughter BW indicated

Table 8. Effect of nutrient intake level on the rate and composition of body tissue retention among 6 slaughter weights in heifers raised at an elevated (E) or restricted (R) nutrient intake level

Item	Treatment[1]												*P*-value		
	100 kg		150 kg		200 kg		250 kg		300 kg		350 kg				
	E	R	E	R	E	R	E	R	E	R	E	R	TRT[1]	BW[2]	Int.[3]
EBW[4] gain, kg	46.3	46.8	78.6	76.5	122.8	127.6	166.7	163.7	214.3[a]	199.7[b]	267.4[a]	248.5[b]	0.01	<0.01	0.01
CP gain, kg	8.5	9.0	14.0	13.9	21.9	22.8	27.1	27.1	32.0	31.6	40.1	38.7	0.76	<0.01	0.61
Fat gain, kg	6.1	5.0	11.1	11.4	17.7	20.1	31.9	27.7	51.0[a]	35.8[b]	67.6[a]	49.7[b]	<0.01	<0.01	<0.01
Ash gain, kg	1.92	2.20	3.40	3.73	5.40	5.54	6.65	7.16	7.81	8.19	11.29	11.00	0.26	<0.01	0.89
H_2O gain, kg	30.5	31.5	50.9	48.4	78.7	80.1	102.0	102.6	120.2	120.9	145.2	145.9	0.79	<0.01	0.94
CP gain, g/d	112.5	84.3	116.2	79.2	128.6	92.1	122.5	89.3	117.9	83.8	121.1	82.4	<0.01	<0.01	0.57
Fat gain, g/d	84.5[a]	47.4[b]	92.9[a]	67.4[b]	104.9[a]	82.9[b]	144.2[a]	91.4[b]	188.2[a]	95.2[b]	204.5[a]	106.8[b]	<0.01	<0.01	<0.01
Ash gain, g/d	26.9	21.4	28.2	20.5	31.5	21.9	29.9	23.3	28.8	22.3	34.5	24.1	<0.01	0.01	0.23
H_2O gain, g/d	414.7	301.6	427.0	279.1	465.9	325.8	462.2	339.6	440.2	321.6	437.2	313.1	<0.01	<0.01	0.46
CP gain, % of EBG[5]	17.5	18.4	17.4	17.7	17.5	17.6	16.1	16.4	15.5	16.4	15.4	16.0	<0.01	<0.01	0.26
Fat gain, % of EBG	13.1[a]	10.5[b]	14.1	15.0	14.5	15.9	19.1[a]	17.0[b]	23.8[a]	18.0[b]	25.2[a]	19.9[b]	<0.01	<0.01	<0.01
Ash gain, % of EBG	4.23	4.70	4.24	4.74	4.32	4.26	3.94	4.29	3.71	4.17	4.29	4.52	<0.01	0.21	0.78
H_2O gain, % of EBG	65.1	66.3	64.3	62.7	63.7	62.4	60.9[b]	62.4[a]	57.1[b]	61.5[a]	55.2[b]	59.6[a]	<0.01	<0.01	<0.01

[a,b]Means with uncommon superscripts within a common slaughter weight are different at $P < 0.05$.
[1]Treatment, R = restricted nutrient intake, E = elevated nutrient intake.
[2]Slaughter weight.
[3]Interaction of TRT and SW.
[4]Empty BW.
[5]EBW gain.

that, except at 150 and 200 kg, the E-heifers deposited a greater proportion of the EBG as fat than R-heifers. This interaction also indicated that the EBG of E-heifers had a lower water content than that of R-heifers at the 3 heaviest slaughtering BW. Retained protein as a percent of EBG was lower in E than in the R-heifers (16.7% vs. 17.0%, respectively, $P = 0.02$) and was also affected by slaughter BW, remaining constant over 17% until 200 kg and decreasing to about 16% at 250 kg to stay unchanged until 350 kg. Retained ash as a percent of EBG was lower for E- than R-heifers (4.1% vs. 4.5%, respectively, $P < 0.01$) regardless of BW.

Absolute and daily RE was greater for E- than R-heifers ($P < 0.01$, Table 9). This divergence between heifers on the 2 treatments was most pronounced at the heaviest slaughter BW. Except at 150 and 200 kg, the E-heifers also retained more energy per unit of EBW than R-heifers ($P < 0.01$). These differences in total RE reflected the differences detected in RE as fat between levels of nutrient intake and their interaction with slaughter BW. Both E- and R-heifers retained a similar amount of RE as protein ($P = 0.32$). However, as a percentage of total RE, E-heifers had less RE as protein than R-heifers at 100, 300, and 350 kg ($P < 0.01$). The RE as protein per kilogram of EBW was affected by slaughter BW ($P < 0.01$), remaining at about 0.94 Mcal/kg to 200 kg, decreasing to 0.86 Mcal/kg at 250 kg, and remaining relatively constant thereafter.

In this study, there was no effect of treatment on the amount of FFM gained, in the proportion of protein and water in the FFM gain (**FFG**), nor in the energy density of the retained FFM (Supplemental Table S2, see Notes). However, an interaction was found between the level of nutrient intake and slaughter BW on FFG as a percentage of the EBG, with lower values in E- than in R-heifers at 150 and 200 kg. Also, the proportion of ash in the FFG tended to be lower in E- than R-heifers (5.0% vs. 5.3%, respectively, $P = 0.09$).

A total of 17 heifers reached puberty before they were slaughtered: 10 E-heifers and 7 R-heifers (Table 10). For these heifers, age at puberty was influenced by the level of nutrient intake, with E-heifers reaching puberty 107 d earlier than R-heifers. Despite this difference in age at puberty, BW at puberty did not differ between treatments. For those reaching puberty before slaughter, the total number of progesterone peaks between puberty and slaughter did not vary between treatments.

DISCUSSION

Previous studies have evaluated the effect of nutrient intake on the composition of growth of Holsteins preweaning, right after weaning, or during the ruminant phase up to puberty. To our knowledge, this is the first

Table 9. Effect of nutrient intake level on deposition of retained energy (RE) among 6 slaughter weights in heifers raised at an elevated (E) or restricted (R) level of nutrient intake starting at 44 kg BW

Item	Treatment												*P*-value		
	100 kg		150 kg		200 kg		250 kg		300 kg		350 kg				
	E	R	E	R	E	R	E	R	E	R	E	R	TRT[1]	BW[2]	Int.[3]
RE,[4] Mcal	100.3	92.3	175.9	177.7	279.6	306.8	441.6	401.9	655.8^{a}	511.9^{b}	854.8^{a}	679.6^{b}	<0.01	<0.01	<0.01
RE as CP, Mcal	45.2	48.0	74.7	73.8	116.4	121.1	144.4	144.0	170.5	168.2	213.6	206.1	0.77	<0.01	0.61
RE as fat, Mcal	57.2	46.7	103.9	106.6	165.9	188.3	298.4	259.0	477.2^{a}	335.6^{b}	633.1^{a}	465.4^{b}	<0.01	<0.01	<0.01
RE, Mcal/d	1.39^{a}	0.89^{b}	1.48^{a}	1.05^{b}	1.66^{a}	1.26^{b}	2.00^{a}	1.33^{b}	2.40^{a}	1.35^{b}	2.57^{a}	1.45^{b}	<0.01	<0.01	<0.01
RE as CP, Mcal/d	0.60	0.45	0.62	0.42	0.68	0.49	0.65	0.48	0.63	0.45	0.64	0.44	<0.01	<0.01	0.57
RE as fat, Mcal/d	0.79^{a}	0.44^{b}	0.87^{a}	0.63^{b}	0.98^{a}	0.78^{b}	1.35^{a}	0.86^{b}	1.76^{a}	0.89^{b}	1.92^{a}	1.00^{b}	<0.01	<0.01	<0.01
RE/EBG,[5] Mcal/kg	2.16^{a}	1.97^{b}	2.25	2.34	2.30	2.42	2.65^{a}	2.47^{b}	3.04^{a}	2.55^{b}	3.17^{a}	2.70^{b}	<0.01	<0.01	<0.01
RE as CP/EBG, Mcal/kg	0.93	0.98	0.93	0.94	0.93	0.93	0.86	0.87	0.82	0.87	0.82	0.85	<0.01	<0.01	0.26
RE as fat/EBG, Mcal/kg	1.23^{a}	0.98^{b}	1.32	1.40	1.36	1.49	1.79^{a}	1.59^{b}	2.22^{a}	1.68^{b}	2.36^{a}	1.86^{b}	<0.01	<0.01	<0.01
RE as CP in total RE, %	43.2^{b}	50.2^{a}	41.4	40.8	40.8	38.8	32.3	35.2	27.3^{b}	34.1^{a}	26.1^{b}	31.5^{a}	<0.01	<0.01	<0.01
RE as fat in total RE, %	56.8^{a}	49.8^{b}	58.6	59.2	59.2	61.2	67.7	64.8	72.7^{a}	65.9^{b}	73.9^{a}	68.5^{b}	<0.01	<0.01	<0.01

[a,b]Means with uncommon superscripts within a common slaughter weight are different at $P < 0.05$.

[1]Treatment, R = restricted nutrient intake, E = elevated nutrient intake.

[2]Slaughter weight.

[3]Interaction of TRT and SW.

[4]Calculated retained tissue energy based on retained fat and protein, and their respective combustion values reported by Blaxter and Rook (1953; 9,367 kcal/kg of fat and 5,322 kcal/kg of protein).

[5]Empty body gain.

Table 10. Age, BW, hip height (HH), and number of ovulations of heifers achieving puberty before slaughter and raised at an elevated (E) or restricted (R) nutrient intake level starting at 44 kg BW

Item	Treatment[1]		SEM	*P*-value
	E	R		
Heifers reaching puberty	10	7	—	—
Age at puberty, d	246^{a}	353^{b}	9.7	<0.01
BW at puberty, kg	270	288	8.4	0.09
HH at puberty, cm	124^{a}	128^{b}	1.6	0.03
Number of ovulations[2]	3.3	4.6	0.92	0.28

[a,b]Means with uncommon superscripts are different at $P < 0.05$.

[1]Treatment, R = restricted nutrient intake; E = elevated nutrient intake.

[2]Number of progesterone peaks >1.0 ng/mL between puberty and slaughter.

study to describe the effects of 2 levels of nutrient intake, initiated early in life, on growth characteristics, efficiency, and composition in Holstein heifers at different stages of development, from weaning through puberty to approximately one-half of mature BW. As expected, BW and HH gains increased with higher nutrient intake. Per the study design, heifers assigned to a common slaughter BW were indeed slaughtered at the same live BW; however, it resulted in E-heifers being younger than R-heifers at slaughter. When assessed at a common BW, E-heifers were consistently shorter at the hips than R-heifers after 250 kg. At a common age, however, E-heifers were taller in the hips than R-heifers (data not shown), because of a more rapid HH growth rate that became apparent by approximately 15 wk of age; however, after approximately 40 wk HH growth in E-heifers tapered off while HH growth in R-heifers continued increasing, resulting in similar HH between the 2 treatment groups. The HH at approximately 10 mo and the rate of HH gain observed here during the 3 to 10 mo period were similar to those reported by Moallem et al. (2004a) for Holstein heifers. The observation that HH growth rate declines as BW increases is consistent with that of Heinrichs and Hargrove (1987).

The estimated initial EBW and composition of the empty body for each heifer were calculated based on the composition of the baseline heifers corresponding to their treatment. In the baseline animals, EBW ranged from 93.8% to 95.1% of pretransport BW. This is consistent with the 95.0% and 96.0% previously reported by our laboratory for similarly sized Holstein heifers (Diaz et al., 2001; Tikofsky et al., 2001) and the ratio of EBW to full BW (**FBW**) used by NASEM (2021). Similarly, the composition of the initial EBW closely resembles that of the baseline groups reported in the studies from our group mentioned above.

For the rest of the heifers slaughtered at different target BW, the amount of BW loss during transport (i.e., transport shrink) from the farm to the abattoir, expressed as a percentage of pretransport BW, was unaffected by BW or level of nutrient intake, averaging 3.96% of BW. Although the post-transport BW from the current study is not equivalent to the shrunk BW described by the CNCPS (Fox et al., 2004), the dairy NRC (2001), and other systems, its relationship to the original BW is almost identical (96% of pretransport and full BW, respectively). However, the behavior observed in the percentage of GIT fill relative to pretransport BW (varying between 14.1% and 22.5%), affected the percentage of EBW in the FBW and that of EBG in the ADG, being all influenced by the level of nutrient intake, BW at slaughter and their interaction. Others have also found differences in the contribution of the GIT contents to the full BW of growing heifers due to the nature of the diet (Waldo et al., 1997) or the stage of growth (Moallem et al., 2004b; Stamey Lanier et al., 2021). Furthermore, Anrique (1976) reported that growing cattle fed ad libitum had about 4% less GIT fill relative to their BW than their counterparts fed restricted amounts of the same diet. Because in the present study the level of nutrient intake altered the contribution of the GIT contents to the FBW only after 250 kg, it is plausible that the greater fat accumulation observed in E-heifers from this point on, accompanied by a relatively constant GIT growth, might have also diluted the contribution of the gut and its contents relative to the FBW compared with R-heifers. In fact, Lofgreen et al. (1962) found that fatness was negatively related to relative gut fill, independent of BW, in beef steers.

The ranges of the contribution of EBW to FBW (between 77.5% and 85.9%) and of EBG to ADG (between 70.5% and 84.9%) observed in the current study indicate that the constants of 85.5%, to convert FBW to EBW, and of 85% to 96%, to convert ADG to EBG, used by different growth models (CNCPS, Fox et al., 2004; NASEM, 2021) oversimplify the dynamics observed in these ratios over the course of development and related to the level of nutrient intake. More precisely, although current conversion constants seem adequate for E-heifers over 300 kg, these overestimate the EBG and EBW of R-heifers and of those lighter than 250 kg. The above-mentioned conversion constants could be traced back to the beef NRC (1984). Despite acknowledging the considerable variation of the contribution of GIT fill to the live BW, that committee and more recent ones (NASEM, 2016, 2021) decided to adopt a simple approach using average conversion ratios recognizing they could lead to incorrect estimates depending on the characteristics of the diet or the growth phase. The estimation of EBW and EBG is fundamental for predicting nutrient requirements and the expected performance of growing heifers. Therefore, a more dynamic approach to estimating these fractions is needed. Williams et al. (1992) proposed one model and evaluated others that might be applicable to growing dairy heifers.

Regardless of the level of nutrient intake, the evolution of the contribution of GIT fill to the BW depicted in Supplemental Figure S1A suggests that the hypertrophy of the GIT, triggered by the marked increase of solid feed during weaning, continued until approximately 150 kg, roughly twice the BW at weaning. At this point, the proportion of GIT fill and EBW in the FBW reached the highest and lowest contributions, respectively. Greater intake and dietary fiber levels, 2 concurrent processes during and after weaning, have been identified as increasing ruminal and intestinal mass (Baldwin et al., 2004). At 100 kg, heifers were still in the process of reaching intake levels relative to BW similar to those observed in subsequent slaughter groups. Because in the present study the GIT was combined with other viscera and blood, its evolution during development could not be appreciated by analyzing the BO fraction alone. However, the inverse relationship between the GIT contribution to the EBW and the EBW:FBW ratio has been observed in other studies investigating the body composition of Holsteins prior and following weaning (Moallem et al., 2004b; Stamey Lanier et al., 2021). In fact, as highlighted by Stamey Lanier et al. (2021), this dynamic affects the EBG:ADG ratio postweaning and, in turn, the nutrient requirement predictions during this phase; here, the EBG:ADG ratio relative to that of EBW:FBW was 90.7% until 150 kg, and then increased from 95.6% at 200 kg to 98% at 350 kg.

Comprehensively, the data related to the composition of the EBW confirms the observation that as growth progresses, fat is retained at a faster rate than the other components and that fat deposition is influenced by level of nutrient intake, particularly after about one-third of the mature BW.

In the current study, E-heifers had greater EBF mass, EBF per unit of EBW and fat gain (absolute and relative to EBG) compared with R-heifers, but generally over 200 kg. Equalizing the allometric equations describing the relationship between the mass of EBF and that of EBW for both treatments and solving for EBW, the EBW at which functions crossed was estimated to be 149 kg, corresponding to 182 kg of FBW or to 28% of the mature BW. As depicted in Figure 1A, before this point EBF masses overlapped for both groups of heifers, but from there after EBF increased at a faster rate relative to the increase of EBW in E- than R-heifers. This is in line with the greater influence of greater nutrient intake on body composition starting from 26% to 40% of the mature BW described by NASEM (2016). These observations, that increasing nutrient intake increases the proportion of fat in EBW and retained EBW, are consistent with those reported by Fortin et al. (1980) and Waldo et al. (1997).

However, the data from the current study confirms that the protein content of the fat-free gain remains unaltered regardless of the level of nutrient intake or BW.

Given the capability to capture BW growth without the confounding effect of the contents in the GIT, the EBG was used to calculate feed efficiency (as unit of EBG per unit of DMI) in the present study. As observed, it is well recognized that feed efficiency decreases as cattle mature (Thonney et al., 1981; Honig et al., 2022). Increased growth and feed efficiency resulting from increased nutrient intake have been observed by our group in milk-fed calves (Diaz et al., 2001) and by others in ruminants, regardless of the type of diet (McLeod and Baldwin, 2000). The results obtained here reinforce these benefits from an enhanced level of nutrient intake started almost at birth and continued almost to puberty. The fact that the difference in feed efficiency between the E- and R-heifer groups disappeared after 250 kg, despite the greater EBG to ADG ratio in the E-heifers, might be related to the greater fat content in the EBG observed in E- respective to R-heifers at the heavier slaughter groups. Adipose tissue has a greater energy density but is lighter than muscular and structural tissue, as it contains less water.

The description of the body fractions and their contribution to the whole EBW and its chemical components provided further insight into the changes on body composition due to level of nutrient intake as heifers mature. The greater contribution of the BO fraction to the EBW observed in E-heifers, could be attributed to the mass increase of the GIT and liver expected with the greater DM and nutrient intake from relatively fibrous diets (McLeod and Baldwin, 2000). The contribution of blood to EBW is suspected to not be part of this change, as this fraction has been reported to be refractory to the level of nutrient intake in growing ruminants (Honig et al., 2022). Also of interest, although EBW fatness was not altered by level of nutrient intake until 200 kg, the compartmentalization of the EBF between body fractions differed across slaughter weights, with E-heifers accumulating a greater proportion of the EBF in the BO fraction. Because the fat concentration of the BO was unaffected by level of nutrient intake until 200 kg (data not shown), the greater proportion of EBF coming from BO fat in E-heifers at lighter slaughter groups could be attributed to the greater contribution of the BO fraction to the EBW alone. After 200 kg, the increased fat content in the EBW due to increased nutrient intake was also observed in all body fractions (data not shown) but it occurred in a greater extent in the BO fraction as could be evidenced by the magnitude of the differences between treatments in the contribution of the BO mass (23.4% vs. 21.1% of EBW) and BO fat (30.1% vs. 25.6% EBF) to the EBW and EBF. It is known that dairy breeds compared with beef breeds of cattle accumulate relatively more fat in internal adipose depots and less in subcutaneous fat (Wright and Russel, 1984). These results are in line with the greater final BW and percentage of omental, mesenteric and perirenal adipose depots of nonlactating, nonpregnant Holstein cows fed a high-energy diet compared with those fed a low energy diet (Drackley et al., 2014). Surprisingly, these authors did not find differences in BCS or carcass weight between the 2 groups of cows and concluded that BCS might not be sensitive enough to detect changes in fat reserves over a short period of time. Although BCS was not included in the present analysis, the differences in relative fat accumulation among body fractions described here sustain the weak correlation of BCS with indirect measurements of body fatness found in growing Holstein heifers (Heine et al., 2021). The evolving proportions of fractions composing the EBW observed in the current study as heifers grew, increasing for the CA and decreasing for the HHFT, agree with the findings observed in growing beef bulls (Honig et al., 2022).

At similar EBW, heifers from the current study had more fat and less protein concentration than those at the beginning (160 kg) and end (280 kg) of the feeding period of Waldo et al. (1997) and a similar observation to the 5 mo (90 kg) and 10 mo (230 kg) old heifers of Moallem et al. (2004b). Both of these previous studies represent relatively contemporary Holsteins to those in the present analysis. Indeed, although Moallem et al. (2004b) did not report the corresponding mature BW to their heifers, that estimated from Waldo et al. (1997; 668 kg) was relatively similar to the BW in this study. Therefore, these body composition differences might be associated with differences in previous levels of nutrient intake. For instance, heifers of Moallem et al. (2004b) were raised as calves on 4.5 L/d of a milk replacer, of unknown composition and reconstituted at an unknown rate, and ad libitum starter until weaning at 60 d of age, and on ad libitum starter only until the initiation of treatments at 3 mo of age (70 kg EBW approximately). At this point heifers had similar fat concentration and higher protein and ash content in the EBW than the baseline calves in the present study. Although no detail is given by Waldo et al. (1997) on the previous rearing conditions of the heifers in their study, it is possible that the Waldo heifers were also raised on a considerably lower level of nutrient intake than the R-heifers in this study, resulting in lower EBW fat and greater EBW protein at the initiation of the study.

Consistent with the dynamic of fat deposition observed in the E- relative to R-heifers, those heifers also had more total TE in the EBW, more RE as fat, greater proportion of fat energy in the total RE and more fat energy and total RE per unit of EBG, particularly after 200 kg. Also, the differences in daily fat deposition rates observed between the 2 levels of nutrient intake resulted in greater daily retention rates of total energy and fat energy in E- than

R-heifers. Although the E-heifers retained more of the total RE as fat than R-heifers the total RE as protein were unaffected by level of nutrient intake. The differences between treatments observed at 100 kg in the energy density of the EBG are related to the greater fat content in the EBG in E-heifers, compared with their R-heifer counterparts. These differences were also reflected in the RE as fat per unit of EBG and in the proportion of RE as fat and protein in the total RE. These differences at 100 kg could be attributed to the higher liquid feeding plan provided to E-heifers as they disappeared in the subsequent BW. Diaz et al. (2001) observed increased fat content in the EBG and greater energy density of the gain in calves fed protein sufficient milk replacers to attain 950 g/d of ADG up to 105 kg compared with those growing at 500 g/d.

The portion of total RE as protein decreased from a high of 49.9% and 43.8% of total RE for R- and E-heifers, respectively, at 100 kg to a low of 31.4% and 26.2% of total RE for R- and E-heifers, respectively, at 350 kg. This shift from energy retained as protein to fat with increasing BW and increasing ADG is consistent with the current understanding of body growth (NASEM, 2016; 2021). When the daily RE was regressed against the daily RE coming from fat and protein, their relationships agreed with those described by others indicating that as daily retention of energy increases the proportional contribution of RE as fat increases and that of RE as protein decreases. However, these relationships were affected by level of nutrient intake ($P < 0.01$; Supplemental Figure S2, see Notes). Although the incremental deviations (or slopes) of fat (0.87) and protein energy (0.13) for R-heifers were similar to those reported by Waldo et al. (1997; 0.89 and 0.11, respectively), the RE at which the lines intercepted, indicating equal contribution of fat and protein energy, was lower than that found by Waldo et al. (0.80 vs. 1.18 Mcal/d). In contrast, the slope of fat (0.95) and protein energy (0.05) of E-heifers were higher, but the energy deposition at equal contributions from fat and protein energy was almost identical to that of Waldo et al. (1997; 1.2 Mcal/d). Moallem et al. (2004b) also observed different slopes for fat (0.76) and protein energy (0.24) than Waldo et al. (1997), but almost the same interception point (1.19 Mcal/d). These differences in slopes and intersection points between studies and treatments might reflect those in composition of the gain, which in turn are influenced by rate of gain and the degree of maturity. In their study, Waldo et al. (1997) indicated that discrepancies in the slopes between treatments might indicate protein inadequacy. However, in the present study at the same daily energy retention, E-heifers retained a greater proportion of protein energy and a lower proportion of fat energy comparted to R-heifers, while receiving the same diets, indicating, in fact, that R-heifers might have been limited by energy to adequately use dietary protein toward protein deposition. The partitioning of RE to either protein or fat observed in the 350 kg slaughter group from the current study is consistent with the observations of Waldo et al. (1997). Further, at 300 kg, heifers from the current study retained a similar if not greater portion of RE as protein than those of Moallem et al. (2004b) until 270 kg (27% to 34% vs. 24% of the total RE).

The onset of puberty signals hormone changes that alter the relationship between fat and protein deposition; thus, it is a milestone for understanding repartitioning and compositional changes of the body. It is generally accepted that the onset of puberty is a function of BW, body composition, or both, rather than chronological age. This is demonstrated by the observations that elevated nutrient intake consistently reduces age at puberty but has little influence on BW at puberty (Capuco et al., 1995; Niezen et al., 1996; Radcliff et al., 1997). Accordingly, E-heifers in the current study reached puberty 107 d before R-heifers. Despite this marked difference in age, BW at puberty was not affected by the level of nutrient intake and averaged 280 kg, corresponding to approximately 43% of the mature BW, almost identical to the maturity at puberty reported previously for Holsteins in NY (42%, Reid et al., 1964) and consistent with the reported range of 43% to 55% of mature BW at which other Holstein strains have reached puberty (Steele et al., 2023). As discussed before, despite their faster rate of HH gain, E-heifers were consistently shorter than R-heifers after 250 kg and, therefore, at puberty. The heifers from Radcliff et al. (1997) and Van Amburgh et al. (1998) raised at faster ADG also reached puberty earlier and were shorter than those grown at slower rates, despite having similar BW. Moallem et al. (2004a) discussed that predominantly lean growth, achieved by a greater skeletal growth with modest increase in BW gain, does not influence the onset of puberty as does a faster rate of BW gain, containing a greater proportion of fat. In fact, it has been established that the onset of puberty in heifers depends on the accumulation of an adequate mass of fat (Fantuz et al., 2024). These observations support the model of Frisch (1991), who suggests that minimal thresholds of body development and relative fatness (~17% of BW as fat in women) are required to achieve puberty. This author indicates that both the absolute and relative amounts of fat are important, because the lean and fat mass must be in a particular absolute and relative range, as the female must be big enough and have enough fat reserves to reproduce successfully.

CONCLUSIONS

The present study complements and expands previous investigations on the composition of growing Holsteins. The increase of fat content leading to a dilution of the fat-free mass in the body as BW increases was accentuated by an elevated nutrient intake only after approximately 30% of the heifer's mature BW without impacting lean growth. Up to this point of maturity, an elevated intake of both protein and energy improved growth and feed efficiency. The additional fat accumulation induced by increased nutrient intake was more pronounced in the visceral adipose depots relative to the other body fractions. The faster growth resulting from the elevated nutrient intake led to the expected decreases in age at a common BW, including age at puberty, which was reduced by 107 d. Taking into consideration the evolving contribution of GIT fill and fat to the BW gain during development and due to level of nutrient intake could refine current descriptions of nutrient requirements and expected growth of Holstein heifers up to midway to maturity.

NOTES

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Nonstandard abbreviations used: Adj. = adjusted; BO = blood and organs; BO–MG = blood and organs excluding the MG; CA = carcass; CNCPS = Cornell Net Carbohydrate and Protein System; E = elevated level of nutrient intake; EBF = empty body fat; EBG = empty BW gain; EBP = empty body protein; EBW = empty BW; FBW = full BW; FFG = FFM gain; FFM = fat-free matter; GIT = gastrointestinal tract; HH = hip height; HHFT = head, hide, feet, and tail; MG = mammary gland; MR = milk replacer; R = restricted level of nutrient intake; RE = retained energy; SW = slaughter weight; TE = tissue energy; TRT = treatment (E or R).

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